

NAKUL PANDHARE

M.SC. INFORMATION TECHNOLOGY

Embedded Systems Engineer with major focus on Robotics and Embedded Software development with expertise in Python, C/C++, and ROS 2

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[portfolio-nakulpandhare](https://portfolio-nakulpandhare.com)

PERSONAL DATA

Birthday 23/11/2000
Birthplace Nagpur, India
Nationality Indian
Currently in Stuttgart, Germany
LinkedIn [in/nakul-pandhare](https://www.linkedin.com/in/nakul-pandhare)
Github [@nakulpandhare](https://github.com/nakulpandhare)

Deeply passionate about embedded systems and robotics systems with hands-on experience in Robot Operating System(ROS), Bluetooth Low Energy(BLE) applications, microcontroller/microprocessor programming, and programming languages like Python and C/C++. I possess strong interest in embedded robotics, intelligent connectivity, and autonomous navigation and intend to develop safe, robust, and intelligent embedded solutions



EDUCATION

Since 10/2022 | M.Sc.

University of Stuttgart, Germany

- Focus:** Information Technology (Major: Embedded Systems)

2018 - 2022 | Bachelors of Engineering

G.H. Rasoni College of Engineering, India

- Focus:** Electronics and Telecommunications Engineering

EXPERTISE

- ROS / ROS 2 Integration
- Embedded Systems
- Robotics and Automation Systems
- Connected Devices
- Simulation
- Programming
- Software Development

RELEVANT EXPERIENCE

Since 01/2025 | German Course (B1 in progress)

07/2024 – Present | Andreas Stihl AG & Co. KG, Waiblingen (Germany)

Werkstudent & Praktikant

- Mastered Bluetooth Low Energy (BLE) communication workflows and implemented some of its features on Nordic nRF52/nRF54 platforms for testing applications
- Gained hands-on experience with Zephyr RTOS, Bluetooth Low Energy and embedded development workflows
- Developed and implemented testing pipelines for device pairing, secure data exchange, and OTA update validation primarily using Python and C/C++
- Integrated BLE stack into an Android (Kotlin) application enabling over-the-air firmware updates and end-to-end connectivity testing
- Documented technical procedures, test results, and user guides to support cross-functional engineering

11/2023 – 04/2024 | BEC Robotics GmbH, Pfullingen (Germany)

Werkstudent Roboticist

- Acquired practical ROS 2 development skills and contributed to testing and operation of an in-house autonomous ground vehicle (AGV) system
- Designed and implemented a ROS 2 (C++ based) interface layer to bridge an industrial mobile robot with an external localization and simulation tool
- Conducted feasibility studies and end-to-end tests on navigation pipelines, ensuring complete data exchange between robot and external tools
- Collaborated directly with software and hardware teams to iterate on automation solutions toward production readiness

STRENGTHS

- Adaptability & Learning Agility
- Cross-Functional Collaboration
- Technical Documentation
- Problem-Solving & Feasibility Assessment
- Time & Project Management

IT SKILLS

- Python
- C/C++
- Robot Operating System (ROS)
- Linux
- Bluetooth LE
- MATLAB
- Microsoft office

LANGUAGES

English	■ ■ ■ ■ ■
German (A2)	■ ■ ■ ■ ■
Hindi	■ ■ ■ ■ ■

01/2023 – 09/2023 | ISW Stuttgart (Institute for Control Technology), Stuttgart (Germany)

University Research Assistant (HiWi)

- Programmed and optimized PLC-driven gantry laser scanners using Beckhoff's TwinCAT, enhancing precision in automated inspection workflows
- Assessed multiple compensation algorithms via simulation and hardware tests to improve precision of the gantry laser cutter
- Supported research topics based on the gantry laser cutting and its outcomes into actionable next steps for production deployments

12/2021 – 06/2022 | Technoventor Innovations Pvt. Ltd, Nagpur (India)

Embedded Product Development Intern

- Learned and gained knowledge on microcontroller programming, sensor-actuator integration
- Gained hands-on experience on programming different microcontrollers and microprocessors
- Prototyped automation system for railway-wagon door detection using Arduino and RF communication
- Evaluated feasibility of remote monitoring POC by implementing real-time data streams from track-side

ACADEMIC PROJECTS

University of Stuttgart, Stuttgart (Germany)

Project: Online Dynamic Risk Assessment for Robot Dog Navigation Task

- Designed and implemented a multi-layer real-time risk assessment framework for a mobile robot dog using Python and computer vision, enabling dynamic obstacle detection and path feasibility evaluation
- Integrated object detection with a custom graph-based model to simulate and assess navigation risk, supporting digital twin-like behavior for predictive decision-making in autonomous tasks

University of Stuttgart, Stuttgart (Germany)

Project: Improvement and Extension of Mobile Robot's Risk Assessment Framework

- Contributed to improving path planner used for navigation in the mobile robot's risk estimation framework
- Worked on advancing basic Probabilistic Roadmap Planner to include dynamic obstacles with known trajectory
- Developed and designed the complete concept using Python on the ROS based framework

G.H. Rasoni College of Engineering, Nagpur (India)

Project: Design of Multipurpose Agro System using Swarm Intelligence

- Designed and built an autonomous multipurpose agricultural robot capable of seed sowing, watering, pesticide spraying, and field surveillance by integrating robotics, IoT, and swarm-intelligence-based path planning
- Developed custom mechanical systems (drilling and seed-dispensing mechanisms), embedded firmware on ATmega microcontroller, and complete robot chassis using CAD design and laser-cut fabrication techniques for real-world farm automation

Opandhore

Note: I am eligible to work in Germany, willing to relocate and available to start immediately at your convenience.